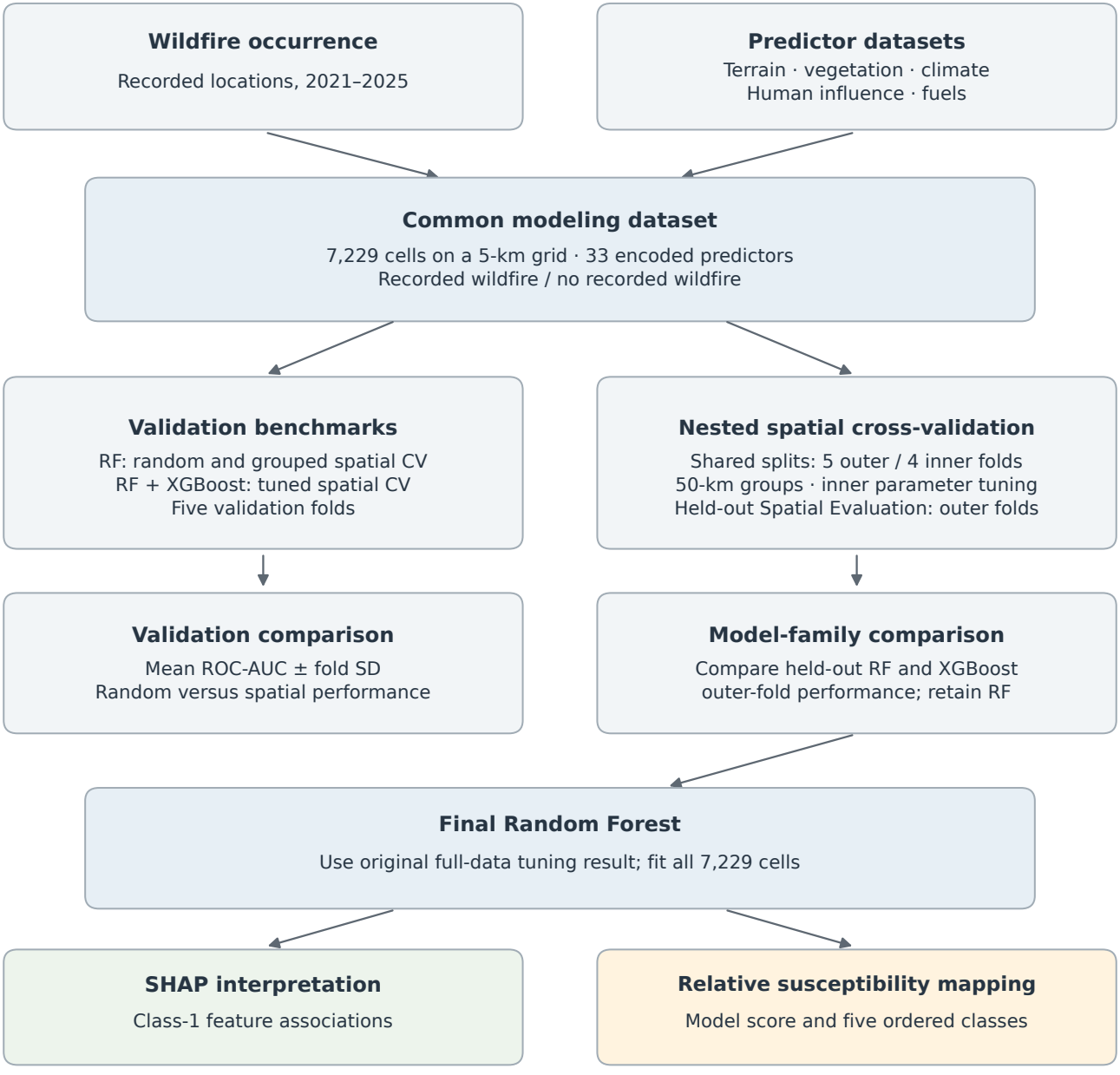
Wildfire occurrence
Recorded locations, 2021–2025
Predictor datasets
Terrain · vegetation · climate
Human influence · fuels
Common modeling dataset
7,229 cells on a 5-km grid · 33 encoded predictors
Recorded wildfire / no recorded wildfire
Validation benchmarks
RF: random and grouped spatial CV
RF + XGBoost: tuned spatial CV
Five validation folds
Nested spatial cross-validation
Shared splits: 5 outer / 4 inner folds
50-km groups · inner parameter tuning
Held-out Spatial Evaluation: outer folds
Validation comparison
Mean ROC-AUC ± fold SD
Random versus spatial performance
Model-family comparison
Compare held-out RF and XGBoost
outer-fold performance; retain RF
Final Random Forest
Use original full-data tuning result; fit all 7,229 cells
SHAP interpretation
Class-1 feature associations
Relative susceptibility mapping
Model score and five ordered classes